

25PY101-S2: Engineering Physics

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References

pre-M1T1

1 References

1.1 Course textbooks

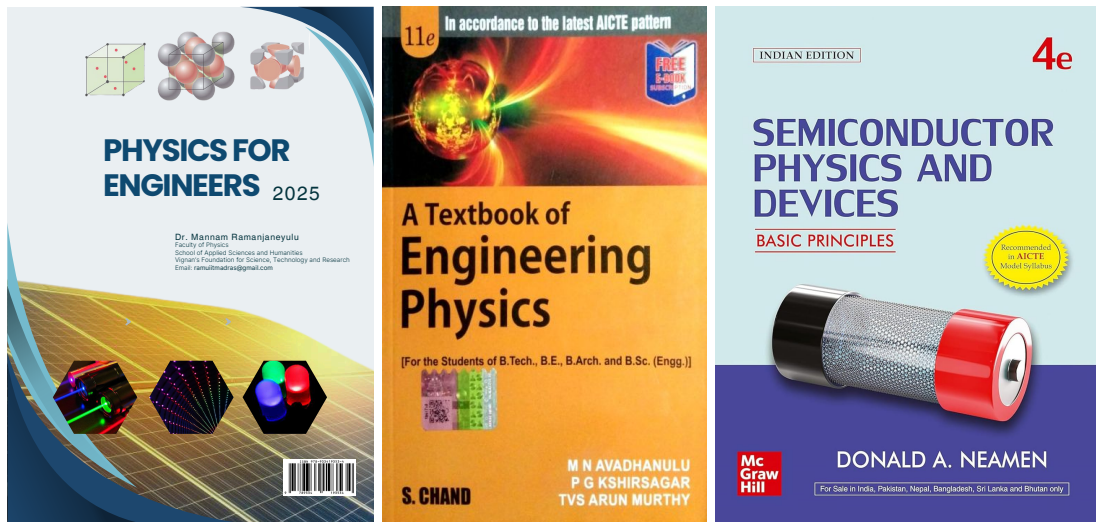


Figure 1: (a) Ramanjaneyulu Mannam (b) Avadhanulu (c) Neamen.

1.2 Reference textbooks

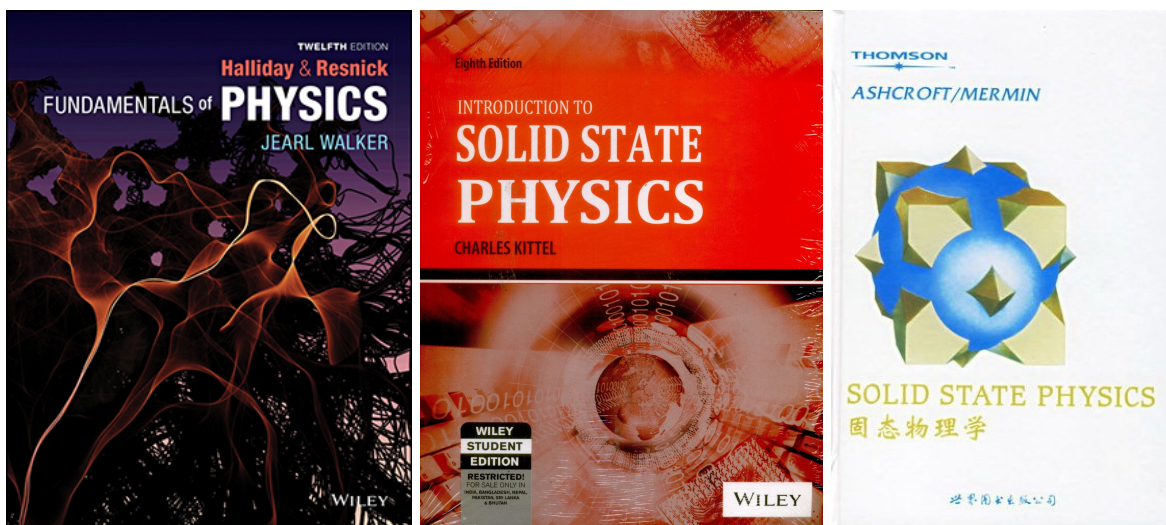


Figure 2: (a) Resnick and Halliday (b) Kittel (c) Ashcroft and Mermin.

2 Syllabus

Module 1	
Unit 1	Introduction to metals, expression for electrical conductivity Introduction to semiconductors–intrinsic and extrinsic Elec. conductivity of s.c. – intrinsic and extrinsic (Quantitatively) Hall effect- applications, Concept of Panchabhuta – five elements
Unit 2	Introduction to Quantum Mechanics, Dual nature of radiation de Broglie’s concept of matter waves, Uncertainty principle Schrödinger’s time-independent wave equation Particle confined in a one-dimensional infinite potential well

Table 1: Syllabus for pre-M1T1.

3 Topics

Topics	Avadhanulu	Neamen
Conductivity problem in metals		
	35.1 – 35.9 Classical Free Electron Theory	
	35-13 – 35.14 Drawbacks of CFET	
Conductivity problem in semiconductors		
	37.1 Introduction	1.1 Semiconductor materials
	37.2 Crystal structure	1.2 – 1.4 Diamond structure
	37.3 Intrinsic semiconductor	
	37.5 Holes	
	37.6 Generation and Recombination	
	37.7 Intrinsic conductivity	
	37.15 Extrinsic semiconductor	
	37.27 Compound Semiconductors	
Hall effect		
	37.28 Hall effect	
Early quantum experiments		
	25.1 – 25.4 Bohr's atomic model	2.0 – 2.1 Principles of quantum mechanics
	24.3 Black body radiation	
	24.4 Planck's quantum hypothesis	
	24.6 Einstein's photo-electric effect	
	26.6 Davisson and Germer experiment	
Dual nature of radiation		
	24.2 Wave picture of radiation	
	3.18 Young's double slit experiment	
	24.5 Particle picture of radiation	
	24.12 Compton effect	
de Broglie's matter waves		
	26.1 – 26.5 de Broglie's matter wave hypothesis	
	26.10 Applications of matter waves	
Heisenberg's uncertainty principle		
	27.1 – 27.7 Implications and applications	
Schrödinger's time-independent wave equation		
	27.8 Wavefunction and probability interpretation	2.2 Schrödinger's wave equation
	27.9 Time dependent and independent versions	

Table 2: Topics and corresponding sections in the course textbooks.

4 Test your understanding

Metals

1. What is the trend of resistivity vs temperature?
2. What is the trend of resistivity vs number density?
3. What is free electron approximation postulate of classical free electron theory (CFET)?
4. What is independent electron approximation postulate of CFET?
5. What is the statistics followed by classical thermodynamics?
6. What are the merits of CFET?
7. What are the demerits of CFET?

Semiconductors

1. What is the trend of resistivity vs temperature for intrinsic semiconductor?
2. If the ionization energy of dopant is 20 meV, the dopants are completely ionized. Explain.
3. What is the condition required for freeze-out of dopants?
4. Under the application of electric field, though the motion of electrons is opposite to the motion of holes, the contribution to the current density from electrons is parallel to contribution from holes. Explain.
5. Though the carrier mobility in metals is less than in semiconductors, conductivity in metals is greater than in semiconductors. Explain.
6. The Hall signal for a copper metal is greater than the Hall signal for a silicon crystal of the same size and shape. Why?
7. The Hall voltage of n-type semiconductor has opposite sign compared to Hall voltage of p-type semiconductor. Why?
8. What are the applications of Hall effect?

Early quantum experiments

Bohr's quantization principle

1. Describe atomic spectra. Differentiate between emission spectra and absorption spectra with suitable examples.
2. State Bohr's quantization principle. Using Bohr's model, explain the origin of spectral lines in hydrogen atom.

Planck's blackbody radiation law

1. State Planck's quantum hypothesis and clearly explain the terms of the corresponding equation.
2. The emission of thermal radiation is given in Figure. 3. In the figure, the variation of energy density $\rho(\lambda, T)$ with wavelength λ is plotted as open circles. The solid line is fit of quantum theory of black body radiation given by Equation. 2 ¹.

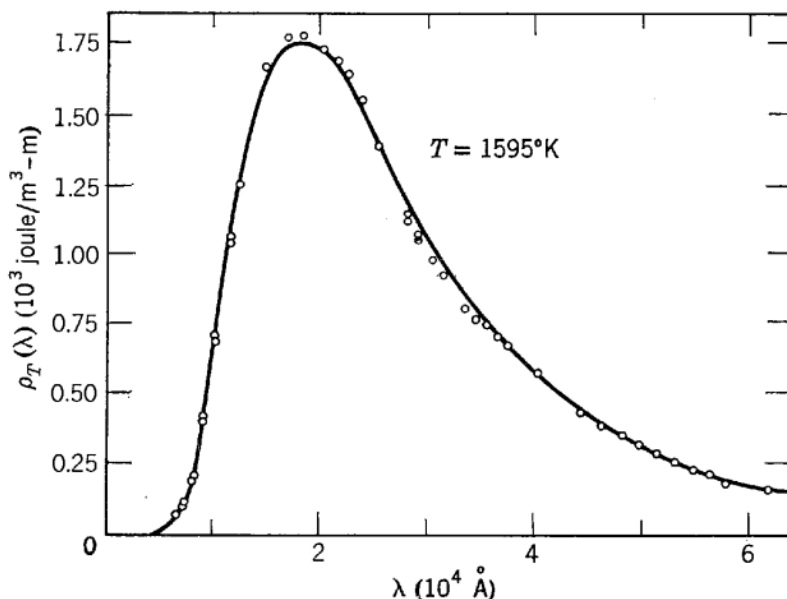


Figure 1-11 Planck's energy density prediction (solid line) compared to the experimental results (circles) for the energy density of a blackbody. The data were reported by Coblentz in 1916 and apply to a temperature of 1595°K. The author remarked in his paper that after drawing the spectral energy curves resulting from his measurements, "owing to eye fatigue it was impossible for months thereafter to give attention to the reduction of the data." The data, when finally reduced, led to a value for Planck's constant of 6.57×10^{-34} joule-sec.

Figure 3: Experimental confirmation of Planck's radiation law. Figure taken from Chapter 1 of Resnick and Eisberg "Quantum Physics of Atoms, Molecules, Solids, Nuclei, and Particles.

Planck conjectured that radiation cannot be absorbed or emitted as any value; instead radiation can only be absorbed or emitted in **quantum/packet** with its energy E and frequency ν related by

$$E = h\nu. \quad (1)$$

The energy density derived by Planck is given by

$$\rho(\nu, T) = \frac{2h\nu^3}{c^2} \frac{1}{\exp\left(\frac{h\nu}{k_B T}\right) - 1}. \quad (2)$$

This equation is the basis of **quantum theory of black body radiation**.

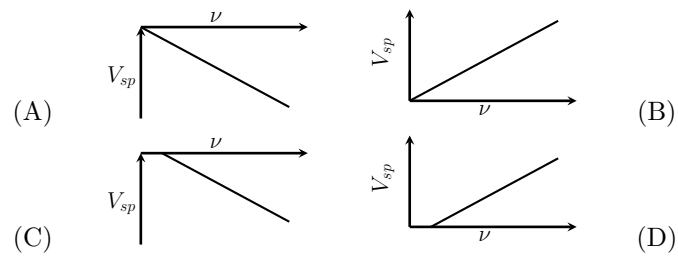
Convince yourself that the radiation at very low wavelength and very high wavelength vanishes.

3. The experiment represented by Figure. 3 was done at 1595 K. If the experiment was done at double the temperature, the wavelength corresponding to the maximum intensity will decrease. Explain.

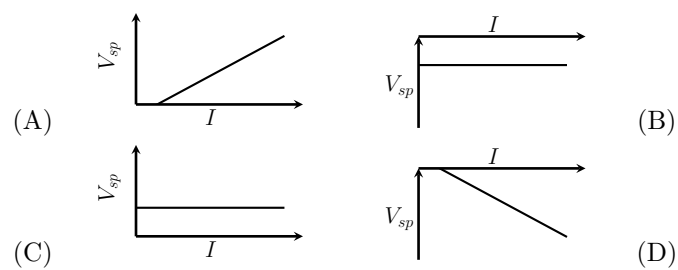
¹Planck presented this work on December 14, 1900 in his seminal paper, "On the Theory of the Energy Distribution Law of the Normal Spectrum". This date is considered as the **birthday of quantum physics**.

Einstein's photo-electric effect

1. Dependence of stopping potential V_{sp} on the photon frequency ν is given by

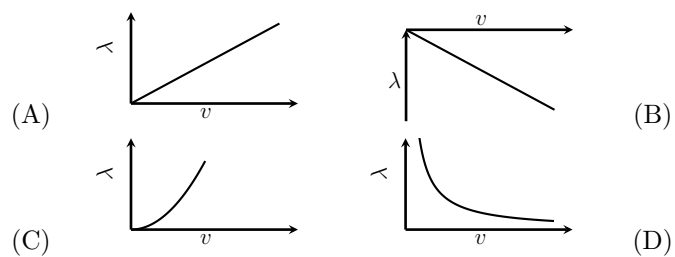


2. Dependence of stopping potential V_{sp} on the intensity I of light of fixed frequency is given by

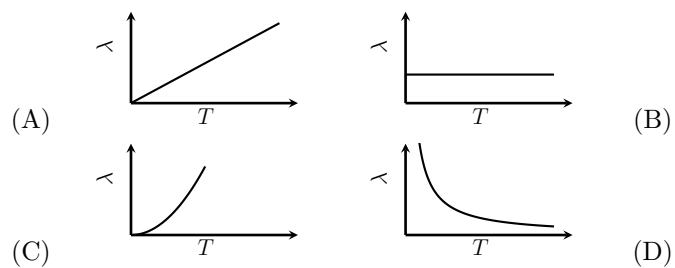


de Broglie's matter waves

1. State de Broglie's matter wave hypothesis and clearly explain the terms of the corresponding equation.
2. Dependence of de Broglie wavelength λ on the velocity v of electron is given by

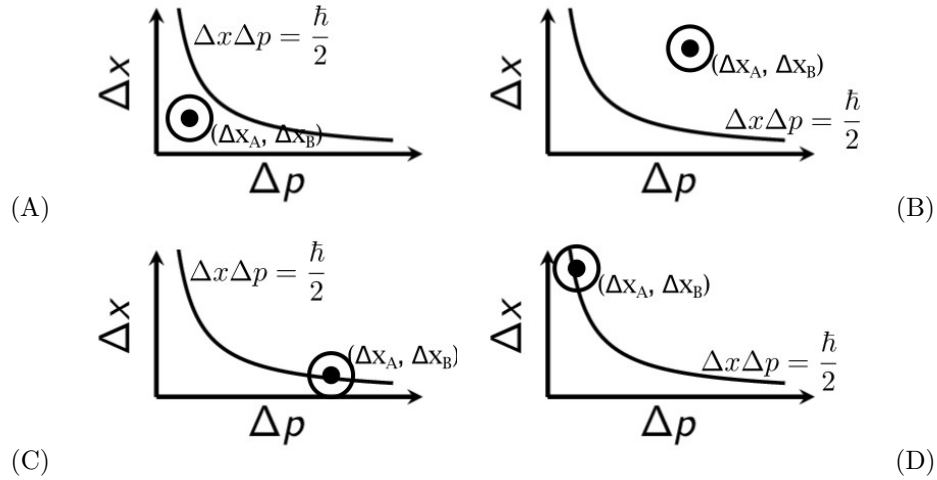


3. Dependence of de Broglie wavelength λ on the temperature T of electron is given by

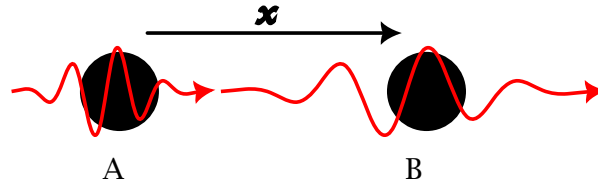


Heisenberg's uncertainty principle

1. State Heisenberg's uncertainty hypothesis and clearly explain the terms of the corresponding **inequality**.
2. In an experiment A, the uncertainty in position Δx_A and uncertainty in momentum Δp_A are measured. which of the following graphs is impossible?



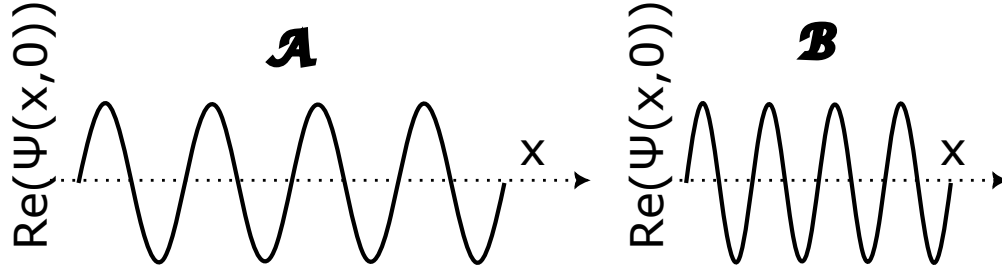
3. Two particles of same mass are traveling along the +x direction.

Figure 4: Choose **True** statement.

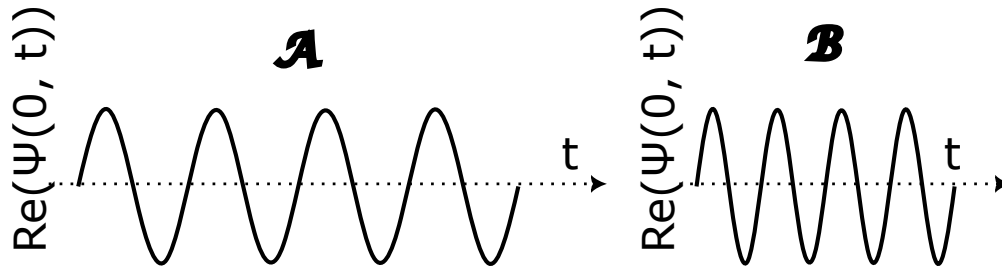
- uncertainty in position of A is greater than uncertainty in position of B
- uncertainty in momentum of A is greater than uncertainty in momentum of B
- uncertainty in position of A is equal to uncertainty in position of B
- uncertainty in momentum of A is equal to uncertainty in momentum of B

Schrödinger's wave equation

1. Write the time dependent and time independent versions of Schrödinger's wave equation and clearly explain the terms.
2. Two particles A and B of same mass have spatial part of their wavefunctions as follows

Figure 5: Choose **False** statement.

- Wavelength of A is greater than wavelength of B
 - Kinetic energy of A is greater than kinetic energy of B
 - Momentum of A is lesser than momentum of B
 - Wavevector of A is lesser than wavevector of B
3. Two particles A and B of same mass have temporal part of their wavefunctions as follows

Figure 6: Choose **True** statement.

- Frequency of A is greater than frequency of B
- Time period of A is lesser than time period of B
- Kinetic energy of A is lesser than kinetic energy of B
- Angular frequency of A is greater than angular frequency of B

5 Technical terms

Metals

- Avogadro's number
- electrical conductivity
- mobility
- relaxation time
- screening
- independent electron approximation
- free electron approximation
- Maxwell-Boltzmann statistics
- thermal motion
- drift motion
- Lorentz force
- Ohm's law
- current density
- free electron density
- drift velocity
- valency

Semiconductors

- intrinsic semiconductor
- elemental semiconductor
- compound semiconductor
- hybridization
- hole
- cohesive energy
- Arrhenius law
- Extrinsic semiconductor
- doping
- dopant
- ionization
- complete ionization
- freeze-out
- Hall voltage
- Hall field
- Hall coefficient

Early quantum experiments

- Quantum
- Photo-electron
- Work function
- Threshold frequency
- Principle of superposition
- matter wave
- uncertainty
- pair of conjugate variables
- wave function
- wave equation
- probability density
- normality condition
- continuity condition
- physicality condition

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